yeast001s — shot list

Every shot in the assembly, in order. **210 shots, 6:51.12**, 23.976p, 1920×1080.

Generated from `picks.json`, so it cannot drift from the timeline. *Sound* is where the audio comes from; *picture* is only filled in where the shot is a split edit and the picture comes from somewhere else.

Sources

Clip	Shots	On screen
`MVI_0040.MOV`	83	3:02.17
`MVI_0019.MOV`	54	1:12.20
`MVI_0041.MOV`	29	0:53.11
`MVI_0038.MOV`	22	0:39.08
`MVI_0021.MOV`	8	0:25.07
`MVI_0037.MOV`	6	0:18.15
`MVI_0045.MOV`	1	0:11.08
`MVI_0043.MOV`	6	0:04.11
`MVI_0036.MOV`	1	0:03.11

Music

Track	In	Out	Level
Audiio_AsTallAsPine_ATAP_TwinLakes	0:00.00	0:29.23	16 dB under the voice
Audiio_SK_Commercial_TwoBigReasons30	6:19.09	6:51.12	16 dB under the voice

Shots

#	At	Dur	Sound	Picture	What it is
1	0:00.00	0.8s	`MVI_0043.MOV @ 13.43`	**Best Yeast Broll.MOV @ 4.00**	hook VO — yeast bubbling in action · sound -0.6 dB
2	0:00.18	0.1s	`MVI_0043.MOV @ 14.31`	**Best Yeast Broll.MOV @ 4.75**	hook VO — yeast bubbling in action (2) · sound -0.6 dB

#	At	Dur	Sound	Picture	What it is
3	0:00.21	1.1s	`MVI_0043.MOV @ 14.56`	**Best Yeast Broll.MOV @ 4.88**	hook VO — yeast bubbling in action (3) · sound -0.6 dB
4	0:01.23	0.8s	`MVI_0043.MOV @ 19.52`	**Katie Dancing.MOV @ 56.81**	hook VO — way too excited about yeast · sound -0.6 dB
5	0:02.17	0.2s	`MVI_0043.MOV @ 20.44`	**Katie Dancing.MOV @ 57.56**	hook VO — way too excited about yeast (2) · sound -0.6 dB
6	0:02.21	1.6s	`MVI_0043.MOV @ 20.85`	**Katie Dancing.MOV @ 57.72**	hook VO — way too excited about yeast (3) · sound -0.6 dB
7	0:04.11	5.3s	`MVI_0040.MOV @ 668.79`	—	hook claim — you don't need sugar, you need time · sound +3.4 dB
8	0:09.18	3.5s	`MVI_0036.MOV @ 35.20`	—	name + promise · sound +2.2 dB
9	0:13.05	4.3s	`MVI_0037.MOV @ 19.02`	—	yeast is really old — Egypt, Asia, Europe · sound +4.4 dB
10	0:17.11	2.0s	`MVI_0037.MOV @ 23.61`	—	yeast is really old — Egypt, Asia, Europe (2) · sound +4.4 dB
11	0:19.11	0.1s	`MVI_0037.MOV @ 26.03`	—	yeast is really old — Egypt, Asia, Europe (3) · sound +4.4 dB
12	0:19.14	2.1s	`MVI_0037.MOV @ 26.32`	—	yeast is really old — Egypt, Asia, Europe (4) · sound +4.4 dB
13	0:21.16	3.5s	`MVI_0037.MOV @ 28.74`	—	yeast is really old — Egypt, Asia, Europe (5) · sound +4.4 dB
14	0:25.04	2.5s	`MVI_0019.MOV @ 123.00`	—	J9 — passed between brewers and bakers; mead, wine, beer · sound -2.4 dB
15	0:27.17	1.4s	`MVI_0019.MOV @ 125.83`	—	J9 — passed between brewers and bakers; mead, wine, beer (2) · sound -2.4 dB
16	0:29.02	0.2s	`MVI_0019.MOV @ 127.42`	—	J9 — passed between brewers and bakers; mead, wine, beer (3) · sound -2.4 dB
17	0:29.06	0.2s	`MVI_0019.MOV @ 127.75`	—	J9 — passed between brewers and bakers; mead, wine, beer (4) · sound -2.4 dB
18	0:29.11	0.1s	`MVI_0019.MOV @ 128.25`	—	J9 — passed between brewers and bakers; mead, wine, beer (5) · sound -2.4 dB

#	At	Dur	Sound	Picture	What it is
19	0:29.14	1.9s	`MVI_0019.MOV @ 128.46`	—	J9 — passed between brewers and bakers; mead, wine, beer (6) · sound -2.4 dB
20	0:31.12	1.5s	`MVI_0019.MOV @ 131.13`	—	J9 — passed between brewers and bakers; mead, wine, beer (7) · sound -2.4 dB
21	0:33.00	1.4s	`MVI_0019.MOV @ 132.92`	—	J9 — passed between brewers and bakers; mead, wine, beer (8) · sound -2.4 dB
22	0:34.10	4.9s	`MVI_0019.MOV @ 134.97`	—	J9 — passed between brewers and bakers; mead, wine, beer (9) · sound -2.4 dB
23	0:39.07	1.5s	`MVI_0019.MOV @ 141.06`	—	J9 — passed between brewers and bakers; mead, wine, beer (10) · sound -2.4 dB
24	0:40.19	0.2s	`MVI_0019.MOV @ 142.89`	—	J9 — passed between brewers and bakers; mead, wine, beer (11) · sound -2.4 dB
25	0:40.23	2.3s	`MVI_0019.MOV @ 143.18`	—	J9 — passed between brewers and bakers; mead, wine, beer (12) · sound -2.4 dB
26	0:43.05	0.1s	`MVI_0019.MOV @ 146.15`	—	J9 — passed between brewers and bakers; mead, wine, beer (13) · sound -2.4 dB
27	0:43.07	1.3s	`MVI_0019.MOV @ 146.35`	—	J9 — passed between brewers and bakers; mead, wine, beer (14) · sound -2.4 dB
28	0:44.15	6.7s	`MVI_0037.MOV @ 213.00`	—	they may not know what they're breeding — dud batch · sound +4.4 dB
29	0:51.07	2.7s	`MVI_0019.MOV @ 170.50`	—	J9 — selective breeding of good yeast, years and years · sound -2.4 dB
30	0:54.00	0.1s	`MVI_0019.MOV @ 173.46`	—	J9 — selective breeding of good yeast, years and years (2) · sound -2.4 dB
31	0:54.03	0.7s	`MVI_0019.MOV @ 174.22`	—	J9 — selective breeding of good yeast, years and years (3) · sound -2.4 dB
32	0:54.19	0.5s	`MVI_0019.MOV @ 175.47`	—	J9 — selective breeding of good yeast, years and years (4) · sound -2.4 dB

#	At	Dur	Sound	Picture	What it is
33	0:55.07	0.9s	`MVI_0019.MOV @ 176.63`	—	J9 — selective breeding of good yeast, years and years (5) · sound -2.4 dB
34	0:56.05	0.2s	`MVI_0019.MOV @ 177.93`	—	J9 — selective breeding of good yeast, years and years (6) · sound -2.4 dB
35	0:56.09	2.0s	`MVI_0019.MOV @ 178.22`	—	J9 — selective breeding of good yeast, years and years (7) · sound -2.4 dB
36	0:58.10	0.6s	`MVI_0019.MOV @ 180.68`	—	J9 — selective breeding of good yeast, years and years (8) · sound -2.4 dB
37	0:59.00	1.8s	`MVI_0019.MOV @ 182.06`	—	J9 — selective breeding of good yeast, years and years (9) · sound -2.4 dB
38	1:00.20	0.2s	`MVI_0019.MOV @ 184.14`	—	J9 — selective breeding of good yeast, years and years (10) · sound -2.4 dB
39	1:01.00	1.6s	`MVI_0019.MOV @ 184.43`	—	J9 — selective breeding of good yeast, years and years (11) · sound -2.4 dB
40	1:02.14	0.5s	`MVI_0038.MOV @ 311.02`	—	until Louis Pasteur — what's being passed on
41	1:03.01	4.4s	`MVI_0038.MOV @ 312.60`	—	until Louis Pasteur — what's being passed on (2)
42	1:07.10	0.2s	`MVI_0038.MOV @ 317.48`	—	until Louis Pasteur — what's being passed on (3)
43	1:07.14	0.5s	`MVI_0038.MOV @ 317.78`	—	until Louis Pasteur — what's being passed on (4)
44	1:08.02	2.1s	`MVI_0038.MOV @ 318.86`	—	until Louis Pasteur — what's being passed on (5)
45	1:10.05	1.6s	`MVI_0038.MOV @ 321.24`	—	until Louis Pasteur — what's being passed on (6)
46	1:11.20	1.4s	`MVI_0038.MOV @ 327.16`	—	has a gooseneck experiment — disproves spontaneous generation
47	1:13.05	2.8s	`MVI_0038.MOV @ 329.75`	—	has a gooseneck experiment — disproves spontaneous generation (2)
48	1:16.00	0.6s	`MVI_0038.MOV @ 332.83`	—	has a gooseneck experiment — disproves spontaneous generation (3)
49	1:16.14	2.1s	`MVI_0038.MOV @ 333.67`	—	has a gooseneck experiment — disproves spontaneous generation (4)

#	At	Dur	Sound	Picture	What it is
50	1:18.16	0.2s	`MVI_0038.MOV @ 336.17`	—	has a gooseneck experiment — disproves spontaneous generation (5)
51	1:18.20	0.9s	`MVI_0038.MOV @ 336.38`	—	has a gooseneck experiment — disproves spontaneous generation (6)
52	1:19.17	4.6s	`MVI_0041.MOV @ 30.49`	—	aside — glass bulb, stretches a neck · sound -0.1 dB
53	1:24.08	0.5s	`MVI_0041.MOV @ 35.62`	—	aside — glass bulb, stretches a neck (2) · sound -0.1 dB
54	1:24.21	0.1s	`MVI_0041.MOV @ 36.62`	—	aside — glass bulb, stretches a neck (3) · sound -0.1 dB
55	1:25.00	2.2s	`MVI_0041.MOV @ 36.91`	—	aside — glass bulb, stretches a neck (4) · sound -0.1 dB
56	1:27.04	0.4s	`MVI_0041.MOV @ 41.12`	—	open here, gooseneck shape — heats it to kill what's inside · sound -0.1 dB
57	1:27.14	0.2s	`MVI_0041.MOV @ 41.79`	—	open here, gooseneck shape — heats it to kill what's inside (2) · sound -0.1 dB
58	1:27.18	2.2s	`MVI_0041.MOV @ 42.00`	—	open here, gooseneck shape — heats it to kill what's inside (3) · sound -0.1 dB
59	1:29.23	0.2s	`MVI_0041.MOV @ 44.63`	—	open here, gooseneck shape — heats it to kill what's inside (4) · sound -0.1 dB
60	1:30.03	3.6s	`MVI_0041.MOV @ 44.88`	—	open here, gooseneck shape — heats it to kill what's inside (5) · sound -0.1 dB
61	1:33.18	1.0s	`MVI_0041.MOV @ 48.88`	—	open here, gooseneck shape — heats it to kill what's inside (6) · sound -0.1 dB
62	1:34.19	2.4s	`MVI_0041.MOV @ 50.30`	—	open here, gooseneck shape — heats it to kill what's inside (7) · sound -0.1 dB
63	1:37.04	0.2s	`MVI_0041.MOV @ 52.97`	—	open here, gooseneck shape — heats it to kill what's inside (8) · sound -0.1 dB
64	1:37.08	0.5s	`MVI_0041.MOV @ 53.26`	—	open here, gooseneck shape — heats it to kill what's inside (9) · sound -0.1 dB
65	1:37.20	1.4s	`MVI_0041.MOV @ 104.73`	—	if spontaneous generation were true, the food breaks down · sound -0.1 dB
66	1:39.06	0.8s	`MVI_0041.MOV @ 106.52`	—	if spontaneous generation were true, the food breaks down (2) · sound -0.1 dB

#	At	Dur	Sound	Picture	What it is
67	1:40.01	2.9s	`MVI_0041.MOV @ 108.19`	—	if spontaneous generation were true, the food breaks down (3) · sound -0.1 dB
68	1:42.22	0.2s	`MVI_0041.MOV @ 111.24`	—	if spontaneous generation were true, the food breaks down (4) · sound -0.1 dB
69	1:43.02	2.9s	`MVI_0041.MOV @ 111.57`	—	if spontaneous generation were true, the food breaks down (5) · sound -0.1 dB
70	1:45.23	1.1s	`MVI_0041.MOV @ 115.16`	—	germ theory — germs get stuck in the downspout · sound -0.1 dB
71	1:47.01	3.4s	`MVI_0041.MOV @ 116.53`	—	germ theory — germs get stuck in the downspout (2) · sound -0.1 dB
72	1:50.10	0.1s	`MVI_0041.MOV @ 120.16`	—	germ theory — germs get stuck in the downspout (3) · sound -0.1 dB
73	1:50.13	1.3s	`MVI_0041.MOV @ 121.70`	—	germ theory — germs get stuck in the downspout (4) · sound -0.1 dB
74	1:51.19	0.2s	`MVI_0041.MOV @ 123.50`	—	germ theory — germs get stuck in the downspout (5) · sound -0.1 dB
75	1:51.23	3.3s	`MVI_0041.MOV @ 123.71`	—	germ theory — germs get stuck in the downspout (6) · sound -0.1 dB
76	1:55.07	0.2s	`MVI_0041.MOV @ 129.38`	—	germ theory — germs get stuck in the downspout (7) · sound -0.1 dB
77	1:55.11	0.9s	`MVI_0041.MOV @ 129.67`	—	germ theory — germs get stuck in the downspout (8) · sound -0.1 dB
78	1:56.09	4.9s	`MVI_0041.MOV @ 134.84`	—	nothing happens to the food — and that's what happens · sound -0.1 dB
79	2:01.07	3.6s	`MVI_0041.MOV @ 146.81`	—	Pasteur proves there is no spontaneous generation · sound -0.1 dB
80	2:04.21	8.3s	`MVI_0041.MOV @ 151.40`	—	Pasteur proves there is no spontaneous generation (2) · sound -0.1 dB
81	2:13.04	2.8s	`MVI_0038.MOV @ 414.62`	—	we meet Saccharomyces cerevisiae
82	2:15.23	4.5s	`MVI_0038.MOV @ 417.67`	—	we meet Saccharomyces cerevisiae (2)
83	2:20.11	0.8s	`MVI_0038.MOV @ 422.42`	—	we meet Saccharomyces cerevisiae (3)

#	At	Dur	Sound	Picture	What it is
84	2:21.05	2.6s	`MVI_0038.MOV @ 423.92`	—	we meet Saccharomyces cerevisiae (4)
85	2:23.20	3.8s	`MVI_0038.MOV @ 427.01`	—	we meet Saccharomyces cerevisiae (5)
86	2:27.15	1.0s	`MVI_0038.MOV @ 431.76`	—	we meet Saccharomyces cerevisiae (6)
87	2:28.14	0.1s	`MVI_0038.MOV @ 432.85`	—	we meet Saccharomyces cerevisiae (7)
88	2:28.17	3.0s	`MVI_0038.MOV @ 433.18`	—	we meet Saccharomyces cerevisiae (8)
89	2:31.16	2.2s	`MVI_0038.MOV @ 436.77`	—	we meet Saccharomyces cerevisiae (9)
90	2:33.21	1.5s	`MVI_0038.MOV @ 439.27`	—	we meet Saccharomyces cerevisiae (10)
91	2:35.09	0.8s	`MVI_0019.MOV @ 279.82`	—	J9 — 90% of all the yeast we use is this one bug · sound -2.4 dB
92	2:36.05	0.2s	`MVI_0019.MOV @ 280.74`	—	J9 — 90% of all the yeast we use is this one bug (2) · sound -2.4 dB
93	2:36.09	2.3s	`MVI_0019.MOV @ 281.24`	—	J9 — 90% of all the yeast we use is this one bug (3) · sound -2.4 dB
94	2:38.15	0.2s	`MVI_0019.MOV @ 283.66`	—	J9 — 90% of all the yeast we use is this one bug (4) · sound -2.4 dB
95	2:38.19	1.6s	`MVI_0019.MOV @ 283.91`	—	J9 — 90% of all the yeast we use is this one bug (5) · sound -2.4 dB
96	2:40.09	1.6s	`MVI_0019.MOV @ 358.11`	**MVI_0026.MOV @ 3.00**	J9 — everywhere, same guy, again and again · sound -2.4 dB
97	2:41.23	1.8s	`MVI_0019.MOV @ 360.07`	**MVI_0026.MOV @ 4.59**	J9 — everywhere, same guy, again and again (2) · sound -2.4 dB
98	2:43.19	0.2s	`MVI_0019.MOV @ 362.40`	**MVI_0026.MOV @ 6.42**	J9 — everywhere, same guy, again and again (3) · sound -2.4 dB
99	2:43.23	0.6s	`MVI_0019.MOV @ 362.70`	**MVI_0026.MOV @ 6.59**	J9 — everywhere, same guy, again and again (4) · sound -2.4 dB
100	2:44.14	0.1s	`MVI_0019.MOV @ 363.82`	**MVI_0026.MOV @ 7.22**	J9 — everywhere, same guy, again and again (5) · sound -2.4 dB

#	At	Dur	Sound	Picture	What it is
101	2:44.17	0.8s	`MVI_0019.MOV @ 364.03`	**MVI_0026.MOV @ 7.34**	J9 — everywhere, same guy, again and again (6) · sound -2.4 dB
102	2:45.11	0.9s	`MVI_0019.MOV @ 366.41`	—	J9 — everywhere, same guy, again and again (7) · sound -2.4 dB
103	2:46.08	0.2s	`MVI_0019.MOV @ 367.53`	—	J9 — everywhere, same guy, again and again (8) · sound -2.4 dB
104	2:46.12	1.8s	`MVI_0019.MOV @ 367.95`	—	J9 — everywhere, same guy, again and again (9) · sound -2.4 dB
105	2:48.07	2.0s	`MVI_0019.MOV @ 370.20`	—	J9 — everywhere, same guy, again and again (10) · sound -2.4 dB
106	2:50.07	1.3s	`MVI_0019.MOV @ 372.54`	—	J9 — everywhere, same guy, again and again (11) · sound -2.4 dB
107	2:51.14	0.4s	`MVI_0019.MOV @ 251.46`	**MVI_0029.MOV @ 2.00**	J9 — mead yeast: S. cerevisiae · sound -2.4 dB
108	2:52.00	1.0s	`MVI_0019.MOV @ 252.46`	**MVI_0029.MOV @ 2.42**	J9 — mead yeast: S. cerevisiae (2) · sound -2.4 dB
109	2:53.01	2.5s	`MVI_0019.MOV @ 253.96`	**MVI_0029.MOV @ 3.46**	J9 — mead yeast: S. cerevisiae (3) · sound -2.4 dB
110	2:55.14	3.0s	`MVI_0019.MOV @ 256.51`	**MVI_0028.MOV @ 1.50**	J9 — instant dry: S. cerevisiae · sound -2.4 dB
111	2:58.13	0.3s	`MVI_0019.MOV @ 259.47`	**MVI_0027.MOV @ 6.01**	J9 — active dry with sugar — guess what · sound -2.4 dB
112	2:58.19	0.2s	`MVI_0019.MOV @ 260.01`	**MVI_0027.MOV @ 6.26**	J9 — active dry with sugar — guess what (2) · sound -2.4 dB
113	2:58.23	3.7s	`MVI_0019.MOV @ 260.22`	**MVI_0027.MOV @ 6.42**	J9 — active dry with sugar — guess what (3) · sound -2.4 dB
114	3:02.16	1.8s	`MVI_0019.MOV @ 264.06`	**MVI_0031.MOV @ 1.00**	J9 — fresh yeast: S. cerevisiae · sound -2.4 dB
115	3:04.12	0.5s	`MVI_0019.MOV @ 265.93`	**MVI_0023.MOV @ 14.01**	J9 — even a pre-ferment is S. cerevisiae, flour and water · sound -2.4 dB
116	3:05.00	3.5s	`MVI_0019.MOV @ 266.68`	**MVI_0023.MOV @ 14.51**	J9 — even a pre-ferment is S. cerevisiae, flour and water (2) · sound -2.4 dB
117	3:08.12	3.8s	`MVI_0019.MOV @ 270.98`	**MVI_0023.MOV @ 18.02**	J9 — even a pre-ferment is S. cerevisiae, flour and water (3) · sound -2.4 dB

#	At	Dur	Sound	Picture	What it is
118	3:12.08	11.3s	`MVI_0045.MOV @ 187.60`	—	someone with a marketing degree — it's all the same · sound +2.3 dB
119	3:23.16	4.3s	`MVI_0040.MOV @ 31.61`	—	why so many names, so many formats? · sound +3.4 dB
120	3:27.23	1.4s	`MVI_0040.MOV @ 45.30`	—	the answer is access and need · sound +3.4 dB
121	3:29.08	3.8s	`MVI_0040.MOV @ 59.43`	—	the farm in Kansas — needs something that won't die off · sound +3.4 dB
122	3:33.02	0.1s	`MVI_0040.MOV @ 63.44`	—	the farm in Kansas — needs something that won't die off (2) · sound +3.4 dB
123	3:33.05	0.6s	`MVI_0040.MOV @ 63.73`	—	the farm in Kansas — needs something that won't die off (3) · sound +3.4 dB
124	3:33.19	6.6s	`MVI_0040.MOV @ 64.90`	—	the farm in Kansas — needs something that won't die off (4) · sound +3.4 dB
125	3:40.10	5.0s	`MVI_0040.MOV @ 71.78`	—	the farm in Kansas — needs something that won't die off (5) · sound +3.4 dB
126	3:45.10	1.0s	`MVI_0040.MOV @ 77.08`	—	the farm in Kansas — needs something that won't die off (6) · sound +3.4 dB
127	3:46.10	0.2s	`MVI_0040.MOV @ 78.33`	—	the farm in Kansas — needs something that won't die off (7) · sound +3.4 dB
128	3:46.14	0.3s	`MVI_0040.MOV @ 78.58`	—	the farm in Kansas — needs something that won't die off (8) · sound +3.4 dB
129	3:46.22	0.8s	`MVI_0040.MOV @ 79.75`	—	the farm in Kansas — needs something that won't die off (9) · sound +3.4 dB
130	3:47.18	0.5s	`MVI_0040.MOV @ 89.76`	—	fresh, dry — active dry and instant · sound +3.4 dB
131	3:48.06	1.5s	`MVI_0040.MOV @ 90.51`	—	fresh, dry — active dry and instant (2) · sound +3.4 dB
132	3:49.18	1.6s	`MVI_0040.MOV @ 92.30`	—	fresh, dry — active dry and instant (3) · sound +3.4 dB
133	3:51.09	5.3s	`MVI_0040.MOV @ 136.09`	—	in the processing, the yeast gets a little damaged · sound +3.4 dB
134	3:56.16	0.1s	`MVI_0040.MOV @ 153.95`	—	it helps dry out the bread a little faster · sound +3.4 dB

#	At	Dur	Sound	Picture	What it is
135	3:56.19	3.0s	`MVI_0040.MOV @ 154.45`	—	it helps dry out the bread a little faster (2) · sound +3.4 dB
136	3:59.18	0.3s	`MVI_0040.MOV @ 217.34`	—	an at-home baker might not care — one loaf at a time · sound +3.4 dB
137	4:00.00	2.9s	`MVI_0040.MOV @ 218.30`	—	an at-home baker might not care — one loaf at a time (2) · sound +3.4 dB
138	4:02.22	0.1s	`MVI_0040.MOV @ 221.43`	—	an at-home baker might not care — one loaf at a time (3) · sound +3.4 dB
139	4:03.01	4.0s	`MVI_0040.MOV @ 221.80`	—	an at-home baker might not care — one loaf at a time (4) · sound +3.4 dB
140	4:07.02	2.0s	`MVI_0040.MOV @ 257.88`	—	not worth driving two hours for a day longer · sound +3.4 dB
141	4:09.03	1.8s	`MVI_0040.MOV @ 260.26`	—	not worth driving two hours for a day longer (2) · sound +3.4 dB
142	4:10.21	3.9s	`MVI_0040.MOV @ 262.43`	—	not worth driving two hours for a day longer (3) · sound +3.4 dB
143	4:14.19	1.5s	`MVI_0040.MOV @ 328.24`	—	mother cell buds a daughter — mitosis, two cells · sound +3.4 dB
144	4:16.06	4.6s	`MVI_0040.MOV @ 330.08`	—	mother cell buds a daughter — mitosis, two cells (2) · sound +3.4 dB
145	4:20.21	2.0s	`MVI_0040.MOV @ 335.46`	—	mother cell buds a daughter — mitosis, two cells (3) · sound +3.4 dB
146	4:22.21	0.2s	`MVI_0040.MOV @ 337.75`	—	mother cell buds a daughter — mitosis, two cells (4) · sound +3.4 dB
147	4:23.01	2.6s	`MVI_0040.MOV @ 338.00`	—	mother cell buds a daughter — mitosis, two cells (5) · sound +3.4 dB
148	4:25.16	5.5s	`MVI_0040.MOV @ 373.54`	—	grows where there are nutrients — sugars and starches · sound +3.4 dB
149	4:31.04	0.3s	`MVI_0040.MOV @ 379.75`	—	grows where there are nutrients — sugars and starches (2) · sound +3.4 dB
150	4:31.12	0.1s	`MVI_0040.MOV @ 380.55`	—	grows where there are nutrients — sugars and starches (3) · sound +3.4 dB
151	4:31.15	1.0s	`MVI_0040.MOV @ 380.71`	—	grows where there are nutrients — sugars and starches (4) · sound +3.4 dB

#	At	Dur	Sound	Picture	What it is
152	4:32.15	3.7s	`MVI_0040.MOV @ 381.92`	—	grows where there are nutrients — sugars and starches (5) · sound +3.4 dB
153	4:36.08	1.6s	`MVI_0040.MOV @ 444.32`	—	simple sugars, zymase → CO_2 and alcohol · sound +3.4 dB
154	4:37.22	1.3s	`MVI_0040.MOV @ 446.65`	—	simple sugars, zymase → CO_2 and alcohol (2) · sound +3.4 dB
155	4:39.04	3.2s	`MVI_0040.MOV @ 448.57`	—	simple sugars, zymase → CO_2 and alcohol (3) · sound +3.4 dB
156	4:42.08	0.2s	`MVI_0040.MOV @ 452.08`	—	simple sugars, zymase → CO_2 and alcohol (4) · sound +3.4 dB
157	4:42.12	1.0s	`MVI_0040.MOV @ 452.37`	—	simple sugars, zymase → CO_2 and alcohol (5) · sound +3.4 dB
158	4:43.11	4.1s	`MVI_0040.MOV @ 466.38`	—	the byproducts we want — alcohol, or the rise we bake off · sound +3.4 dB
159	4:47.13	2.1s	`MVI_0040.MOV @ 470.80`	—	the byproducts we want — alcohol, or the rise we bake off (2) · sound +3.4 dB
160	4:49.16	0.5s	`MVI_0040.MOV @ 473.85`	—	the byproducts we want — alcohol, or the rise we bake off (3) · sound +3.4 dB
161	4:50.03	0.1s	`MVI_0040.MOV @ 475.10`	—	the byproducts we want — alcohol, or the rise we bake off (4) · sound +3.4 dB
162	4:50.06	0.8s	`MVI_0040.MOV @ 475.31`	—	the byproducts we want — alcohol, or the rise we bake off (5) · sound +3.4 dB
163	4:51.01	3.9s	`MVI_0040.MOV @ 476.56`	—	the byproducts we want — alcohol, or the rise we bake off (6) · sound +3.4 dB
164	4:54.22	3.2s	`MVI_0019.MOV @ 713.38`	—	J9 — that's where you get your bubbly wines, like champagne · sound -2.4 dB
165	4:58.02	2.0s	`MVI_0019.MOV @ 716.88`	—	J9 — that's where you get your bubbly wines, like champagne (2) · sound -2.4 dB
166	5:00.01	1.5s	`MVI_0040.MOV @ 492.24`	—	the right environment — a temperature between… · sound +3.4 dB
167	5:01.14	2.3s	`MVI_0040.MOV @ 494.45`	—	the right environment — a temperature between… (2) · sound +3.4 dB
168	5:03.22	1.2s	`MVI_0040.MOV @ 497.04`	—	the right environment — a temperature between… (3) · sound +3.4 dB

#	At	Dur	Sound	Picture	What it is
169	5:05.03	0.2s	`MVI_0040.MOV @ 498.41`	—	the right environment — a temperature between… (4) · sound +3.4 dB
170	5:05.07	5.7s	`MVI_0040.MOV @ 498.71`	—	the right environment — a temperature between… (5) · sound +3.4 dB
171	5:11.00	7.2s	`MVI_0040.MOV @ 505.13`	—	…50 degrees dormant and 140 killed off · sound +3.4 dB
172	5:18.05	0.1s	`MVI_0040.MOV @ 512.72`	—	…50 degrees dormant and 140 killed off (2) · sound +3.4 dB
173	5:18.08	0.3s	`MVI_0040.MOV @ 513.05`	—	…50 degrees dormant and 140 killed off (3) · sound +3.4 dB
174	5:18.14	5.0s	`MVI_0040.MOV @ 518.10`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive · sound +3.4 dB
175	5:23.14	0.4s	`MVI_0040.MOV @ 523.98`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (2) · sound +3.4 dB
176	5:24.00	1.8s	`MVI_0040.MOV @ 524.65`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (3) · sound +3.4 dB
177	5:25.19	0.2s	`MVI_0040.MOV @ 526.86`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (4) · sound +3.4 dB
178	5:25.23	1.5s	`MVI_0040.MOV @ 527.07`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (5) · sound +3.4 dB
179	5:27.11	0.2s	`MVI_0040.MOV @ 528.78`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (6) · sound +3.4 dB
180	5:27.15	0.4s	`MVI_0040.MOV @ 529.03`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (7) · sound +3.4 dB
181	5:28.00	1.1s	`MVI_0040.MOV @ 529.78`	—	the sweet spot — 93 to 100-and-a-couple — grow and thrive (8) · sound +3.4 dB
182	5:29.03	6.6s	`MVI_0040.MOV @ 551.01`	—	salt, fat, spices get in the way — slows proving · sound +3.4 dB
183	5:35.18	0.1s	`MVI_0040.MOV @ 558.89`	—	salt, fat, spices get in the way — slows proving (2) · sound +3.4 dB
184	5:35.20	10.8s	`MVI_0040.MOV @ 559.10`	—	salt, fat, spices get in the way — slows proving (3) · sound +3.4 dB
185	5:46.14	1.4s	`MVI_0040.MOV @ 583.12`	—	sugar speeds it up · sound +3.4 dB

#	At	Dur	Sound	Picture	What it is
186	5:47.23	0.1s	`MVI_0040.MOV @ 584.88`	—	sugar speeds it up (2) · sound +3.4 dB
187	5:48.02	1.6s	`MVI_0040.MOV @ 585.08`	—	sugar speeds it up (3) · sound +3.4 dB
188	5:49.17	7.6s	`MVI_0040.MOV @ 625.12`	—	active dry says add sugar — you don't have to, let it sit · sound +3.4 dB
189	5:57.08	0.1s	`MVI_0040.MOV @ 633.42`	—	active dry says add sugar — you don't have to, let it sit (2) · sound +3.4 dB
190	5:57.10	4.6s	`MVI_0040.MOV @ 633.55`	—	active dry says add sugar — you don't have to, let it sit (3) · sound +3.4 dB
191	6:02.00	2.8s	`MVI_0040.MOV @ 648.19`	**MVI_0025.MOV @ 20.02**	yeast just wants flour and water — the pre-ferment, bubbling · sound +3.4 dB
192	6:04.20	0.2s	`MVI_0040.MOV @ 651.11`	**MVI_0025.MOV @ 22.86**	yeast just wants flour and water — the pre-ferment, bubbling (2) · sound +3.4 dB
193	6:05.00	0.3s	`MVI_0040.MOV @ 651.48`	**MVI_0025.MOV @ 23.02**	yeast just wants flour and water — the pre-ferment, bubbling (3) · sound +3.4 dB
194	6:05.07	3.5s	`MVI_0040.MOV @ 652.23`	**MVI_0025.MOV @ 23.31**	yeast just wants flour and water — the pre-ferment, bubbling (4) · sound +3.4 dB
195	6:08.20	0.2s	`MVI_0040.MOV @ 656.11`	**MVI_0025.MOV @ 26.86**	yeast just wants flour and water — the pre-ferment, bubbling (5) · sound +3.4 dB
196	6:09.00	0.3s	`MVI_0040.MOV @ 656.32`	—	yeast just wants flour and water — the pre-ferment, bubbling (6) · sound +3.4 dB
197	6:09.06	0.1s	`MVI_0040.MOV @ 656.82`	—	yeast just wants flour and water — the pre-ferment, bubbling (7) · sound +3.4 dB
198	6:09.09	0.7s	`MVI_0040.MOV @ 657.03`	—	yeast just wants flour and water — the pre-ferment, bubbling (8) · sound +3.4 dB
199	6:10.02	0.1s	`MVI_0040.MOV @ 657.99`	—	yeast just wants flour and water — the pre-ferment, bubbling (9) · sound +3.4 dB
200	6:10.05	4.9s	`MVI_0040.MOV @ 658.32`	—	yeast just wants flour and water — the pre-ferment, bubbling (10) · sound +3.4 dB
201	6:15.03	3.8s	`MVI_0040.MOV @ 674.88`	—	adding sugar speeds it up, but — sweeter dough · sound +3.4 dB
202	6:18.22	7.3s	`MVI_0040.MOV @ 682.10`	—	if that's not what you want, you don't need sugar · sound +3.4 dB

#	At	Dur	Sound	Picture	What it is
203	6:26.05	1.8s	`MVI_0021.MOV @ 415.87`	—	J9 — use what's available, it's all the same · sound -0.3 dB
204	6:27.23	1.1s	`MVI_0021.MOV @ 486.40`	—	J9 — not a hard and fast rule; the same little bug again and again · sound -0.3 dB
205	6:29.01	0.1s	`MVI_0021.MOV @ 487.82`	—	J9 — not a hard and fast rule; the same little bug again and again (2) · sound -0.3 dB
206	6:29.04	11.1s	`MVI_0021.MOV @ 488.03`	—	J9 — not a hard and fast rule; the same little bug again and again (3) · sound -0.3 dB
207	6:40.06	4.0s	`MVI_0021.MOV @ 500.79`	—	J9 — all the little bug wants is to eat sugars and fart (⚠ channel voice) · sound -0.3 dB
208	6:44.07	0.2s	`MVI_0021.MOV @ 505.34`	—	J9 — all the little bug wants is to eat sugars and fart (⚠ channel voice) (2) · sound -0.3 dB
209	6:44.11	1.1s	`MVI_0021.MOV @ 506.01`	—	J9 — all the little bug wants is to eat sugars and fart (⚠ channel voice) (3) · sound -0.3 dB
210	6:45.14	5.9s	`MVI_0021.MOV @ 514.89`	—	J9 — produce CO_2 and ethanol; a happy life; not that complicated · sound -0.3 dB

What each shot says

1. 0:00.00 — This is yeast bubbling in action.

4. 0:01.23 — And this is me way too excited about yeast.

7. 0:04.11 — You don't actually need sugar to make your yeast grow. You just need time.

8. 0:09.18 — Hi, I'm Katie. I'm a food scientist, and this is my deep dive into yeast.

9. 0:13.05 — Yeast is really old. We've had it around for at least a few thousand years. We've found it in archaeological records in Egypt. We have found it in archaeological records in parts of Asia and in Europe.

14. 0:25.04 — It's being passed between brewers and bakers. It's being collected from the air. It's being collected from outside of grapes and introduced into wine. It's moving around. It's passed from person to person, baker to brewer, brewer to baker. It's in meat. It's in wine. It's in beer. It's in our sourdough starters. It's just all around.

28. 0:44.15 — They may not know what they're breeding, but they're being selective about it. If it's a dud batch, they're not passing it on to their friend.

29. 0:51.07 — And so what we're getting is selective breeding of good yeast, yeast that ferments, that makes the bread bubble, makes the alcohol happen, is being passed around. And that is what we've used for years and years and years.

40. 1:02.14 — It wasn't until we meet Louis Pasteur that we have an inkling of what's making one successful batch work and what's being passed on between these bakers and brewers.

46. 1:11.20 — has a gooseneck experiment and disproves spontaneous generation and says there are microbes, little bugs that we can't see all around the world.

52. 1:19.17 — If you don't know about the Louis Pasteur experiment, what he does is he takes a glass bulb and stretches out a neck on it.

56. 1:27.04 — that is open over here, but it has a gooseneck shape. And he heats it up to kill off any bacteria that's on the inside with some food that he has cooked, that he has put inside of this gooseneck bottle.

65. 1:37.20 — And so if there's any truth to spontaneous generation, the food will just break down. There will be no problem with it doing what food has always done.

70. 1:45.23 — But if the food breaks down because of germ, germ theory, what he was thinking of is germs can't get in through the gooseneck bottle. It falls down. It gets stuck in the downspout right here.

78. 1:56.09 — so nothing will happen to the food. It won't break down. It won't do anything. And that's what happens.

79. 2:01.07 — And so Louis Pasteur proves that there is no spontaneous generation, that there is something in the air, bacteria, yeast, living organisms that are getting into our food and making changes.

81. 2:13.04 — meet our Saccharomyces cerevisiae. And this is the most common form of yeast that we use today. There are other yeasts that are used in sourdoughs, but for the most part, when you get to industrial bakers and you get to at-home bakers and you get to brewers, they're all using the same little bug, Saccharomyces cerevisiae.

91. 2:35.09 — It is like 90% of all the yeast that we use is this one bug.

96. 2:40.09 — all over we use it everywhere from our home bakeries to the brewers down the road to the vineyards in California and France, everywhere, Saccharomyces cerevisiae, same guy, again and again and again.

107. 2:51.14 — I have marketing for yeast for mead. It's Saccharomyces cerevisiae.

110. 2:55.14 — I have instant dry yeast. It's Saccharomyces cerevisiae.

111. 2:58.13 — I have active dry yeast with a little sugar in it. Guess what? Saccharomyces cerevisiae.

114. 3:02.16 — Fresh yeast. Saccharomyces cerevisiae.

115. 3:04.12 — I even have what could be the start of a sourdough starter. Here, it's a pre-ferment. It is yeast, Saccharomyces cerevisiae mixed with flour and water.

118. 3:12.08 — And so someone with a marketing degree, with a business degree is saying, hey, let me sell to you. I know what you need. It's this. But it's really all just the same Saccharomyces cerevisiae.

119. 3:23.16 — Why do we have so many different names for it? Why do we have so many different formats for it?

120. 3:27.23 — The answer is access and need.

121. 3:29.08 — you have the at-home baker in the middle of a farm in Kansas that needs something every once in a while and doesn't want to have to drive two hours back and forth to a grocery store every couple days to get fresh yeast. They need something on hand that they can keep that won't die off on them as frequently.

130. 3:47.18 — So you have fresh yeast and dry yeast and you have active dry and instant.

133. 3:51.09 — But in the processing of the active dry yeast and the instant yeast, the yeast gets a little damaged

134. 3:56.16 — it helps dry out the bread a little bit faster.

136. 3:59.18 — And to be honest, an at-home baker might not care. You might not notice. They consume the bread within a couple days because they're only making one loaf at a time.

140. 4:07.02 — because it might not be worth driving a couple hours to a grocery store to get something to have a bread that's a little bit stretchier that lasts a day longer.

143. 4:14.19 — So you have a mother cell that grows and buds on itself a daughter cell. And when that daughter cell is big and identical to the mother cell, it uses mitosis and splits into two cells.

148. 4:25.16 — yeast will continue to grow and reproduce itself when it's in an environment that has lots of nutrients for it. And those nutrients are sugars and starches. It breaks down starch into sugar.

153. 4:36.08 — the yeast will use simple sugars and break it down with the enzyme zymase into CO2 and alcohols.

158. 4:43.11 — and the CO2 and the alcohol are the byproducts that we're looking for as humans, either in alcohol production or in helping our bread rise with the CO2, giving the gas that we will eventually bake off.

164. 4:54.22 — And you also have your CO2. That's where you get your bubbly wines like your champagne.

166. 5:00.01 — yeast is looking for a right environment for it to thrive and to grow and that environment has lots of sugars and starches and is in a temperature between 50

171. 5:11.00 — degrees where it kind of shuts down and goes dormant and 140 degrees when it's completely killed off

174. 5:18.14 — There's a nice sweet spot right in the middle, about 93 degrees Fahrenheit and 100 plus a couple degrees Fahrenheit. And in that sweet temperature spot, it will grow and thrive.

182. 5:29.03 — so when we add salt and fat and other spices to our bread doughs, what we're doing is we're putting items in the way of the yeast and its food and so it's slowing down the process in which the yeast can digest the sugars that it wants for growing and that will then slow our proving time.

185. 5:46.14 — the availability of sugar will speed up the proofing time.

188. 5:49.17 — this is why in active dry yeast they recommend that you add a little bit of sugar to kind of start the yeast up and going and getting it bubbling you don't have to actually do that you can just let it sit a little longer

191. 6:02.00 — yeast itself just really wants flour and water and you can do that sometimes by getting a pre-ferment. This is what I have going right here and it's already starting to bubble up and it's just flour and water and yeast and the yeast is in a happy spot.

201. 6:15.03 — Adding sugar does speed it up, but then you end up with a sweeter dough,

202. 6:18.22 — and if that is not the final product that you want, you do not actually need to add sugar to your dough to get the yeast to grow.

203. 6:26.05 — Use what's available. It's all the same.

204. 6:27.23 — It can be played with. Yeast is not something that has to be a hard and fast rule. It feels like a mystery, but really all it is is the same little Saccharomyces cerevisiae bug again and again and again and again.

207. 6:40.06 — All the little bug wants is to go and eat sugars and fart.

210. 6:45.14 — produce carbon dioxide and ethanol. It just wants to live a happy life and it's not all that complicated.